

Executive Summary

This project focuses on predicting passenger survival on the Titanic using Machine Learning classification techniques. The main objective of the project is to analyze passenger-related information and build a predictive model capable of determining whether a passenger survived or not during the Titanic disaster. The dataset used for this project was obtained from Kaggle and contains various passenger attributes such as passenger class, gender, age, fare, family details, and embarkation information.

The project began with Exploratory Data Analysis (EDA) to understand the structure of the dataset and identify important patterns affecting passenger survival. Different visualizations and analyses were performed to study survival distribution, passenger demographics, and feature relationships. Data preprocessing techniques were applied to improve model performance and ensure data quality.

A major focus of the project was feature engineering. New features such as passenger title extraction, family size, and cabin presence were created to enhance the predictive capability of the model. Missing values in important columns such as Age and Embarked were handled using appropriate strategies including median and mode imputation. Categorical variables were encoded using Label Encoding, and feature scaling was applied using StandardScaler.

Two classification algorithms were implemented and compared:

- Logistic Regression
- Random Forest Classifier

The dataset was divided into training and testing sets using train-test split methodology. The models were evaluated using classification metrics including accuracy score, precision, recall, F1-score, and confusion matrix analysis. Random Forest Classifier achieved the best performance for predicting passenger survival.

To improve model explainability, feature importance analysis was performed using Random Forest feature importance scores. This helped identify the most influential features contributing to survival prediction.

The trained model was saved using Joblib in `.pkl` format to allow future predictions without retraining the model. This project demonstrates the complete machine learning pipeline including data preprocessing, feature engineering, model training, evaluation, explainability, and deployment preparation. The final deliverables include the Jupyter Notebook, saved model file, screenshots, README documentation, and GitHub repository link.